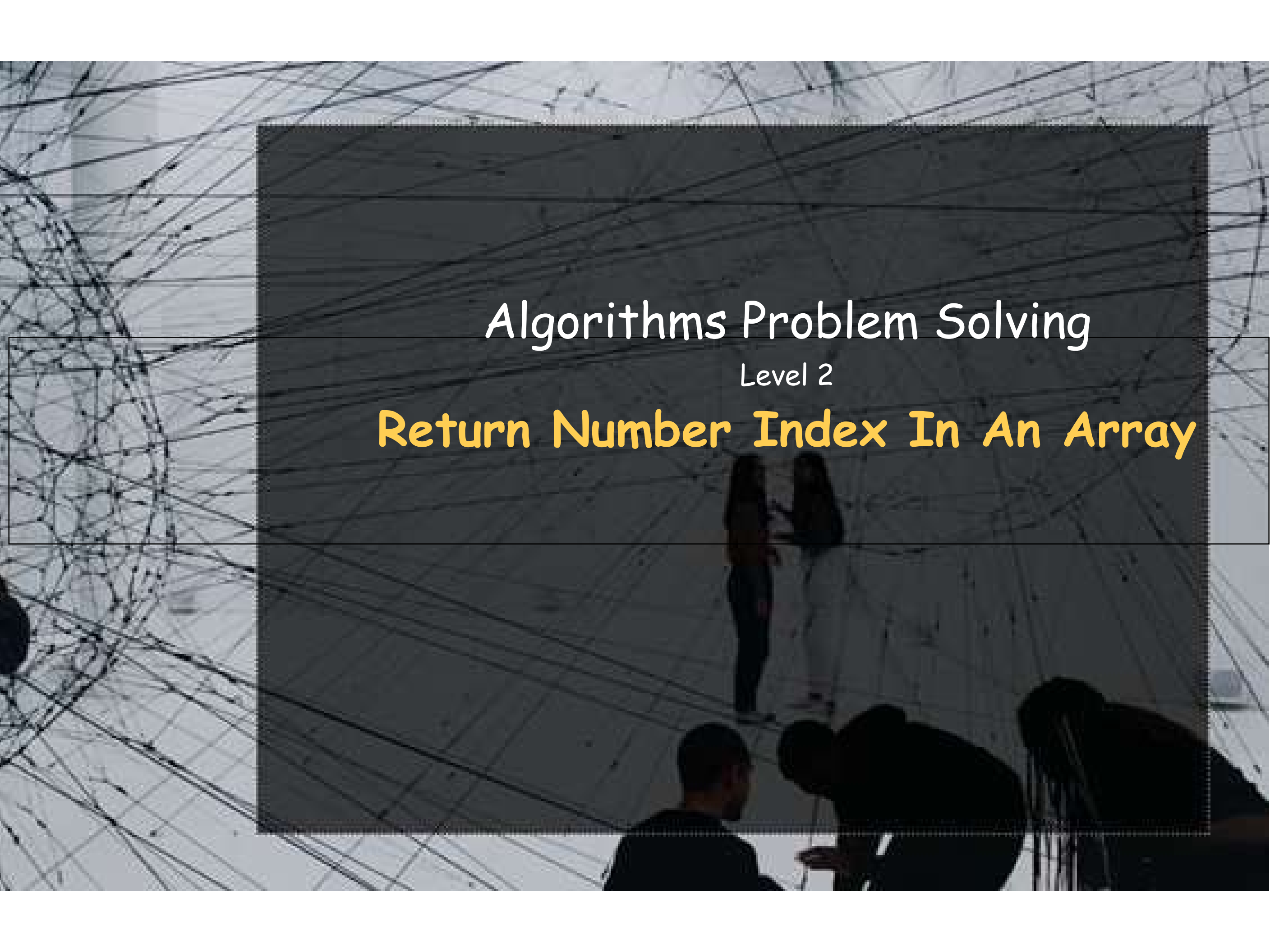


Algorithms Problem Solving

Level 2

Return Number Index In
An Array



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Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It asks the user for a number to search for.
- ◆ It searches for the number inside the array.
- ◆ If the number is found, the function returns its index.
- ◆ If the number is not found, the function returns `-1`.

Explanation

- ◆ ``FindNumberPositionInArray()`` searches through the array from the first element to the last.
- ◆ The loop checks each element using:
 `if (arr[i] == numb)`
- ◆ If the number is found, the function immediately returns its index:
 `return i;`
- ◆ If the loop finishes without finding the number, it returns:
 `return -1;`
- ◆ Index starts from ``0``, while the order starts from ``1``.
- ◆ For example, if the number is found at index ``0``, its order is ``1``.

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `1` to `100`.
3. Print all array elements.
4. Ask the user for the number to search for.
5. Call `FindNumberPositionInArray()`.
6. Compare each array element with the searched number.
7. Return the index if the number is found.
8. Return `-1` if the number is not found.
9. Print the search result.

💡 Example

◆ `Array Elements` :

19 70 18 86 70 9 16 23

◆ Number to search:

19

◆ The first element is `19`.

◆ Its index is `0`.

◆ Its order is `1`.

➡ Result

Number you are looking for is: 19

The number found at position (index): 0

The number found its order: 1

📌 Note

◆ Important function:

```
int FindNumberPositionInArray(int arr[100], int arrayLength, int numb)
```

◆ `arr` is the array to search.

◆ `arrayLength` tells the function how many elements to check.

◆ `numb` is the number we are looking for.

◆ `return i` means the number was found at index `i`.

◆ `return -1` means the number was not found.

◆ If the searched number appears more than once, the function returns the index of the first occurrence.

⊘ Warning

- ◆ The array has a maximum size of `100`, so `arrayLength` should not be greater than `100`.
- ◆ `ReadNumber()` only checks that the number is greater than `0`; it does not enforce the maximum of `100` for the array length.
- ◆ `cmath` and `corecrt\_math\_defines.h` are not needed in this program.
- ◆ `srand()` and `time()` should ideally have their proper headers included explicitly.
- ◆ Because the array contains random numbers, the same number can appear more than once.



Comparison

▨ `return i`

- ✓ Used when the number is found.
- ✓ Returns the index where the number was found.
- ✓ Example: `19` at index `0` → returns `0`.

▤ `return -1`

- ✓ Used when the number is not found.
- ✓ Indicates that there is no valid index.
- ✓ Example: searching for `50` when it does not exist → returns `-1`.

✦ Summary

- ✓ Fill the array with random numbers.
- ✓ Ask the user for a number to search for.
- ✓ Check every array element.
- ✓ Return the index when the number is found.
- ✓ Return `-1` when the number is not found.
- ✓ Remember: array index starts from `0`, while order starts from `1`.

Thank you 😊